

## Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

### 10.01 Interpreting graphs (booklet p. 3)

#### WE DO examples (teacher)

1. Ex 1a: Gold Coast
2. Ex 1b: Uluru
3. Ex 2a: 15–24 years
4. Ex 2b: 12%
5. Ex 3a: Jan and Feb, 18°C
6. Ex 3b: it falls steadily to a low in July

#### Easy (drill)

- |         |            |             |
|---------|------------|-------------|
| 1. 47.5 | 10. 18     | 19. 48      |
| 2. 32.5 | 11. 130    | 20. 2       |
| 3. 11   | 12. 7      | 21. 27°C    |
| 4. 170  | 13. 13     | 22. 16°C    |
| 5. 150  | 14. 8      | 23. January |
| 6. 15   | 15. 6      | 24. June    |
| 7. 37.5 | 16. 15     | 25. 11°C    |
| 8. 55   | 17. 6      | 26. 16°C    |
| 9. 85   | 18. Tennis |             |

#### Medium (drill)

- |          |          |         |
|----------|----------|---------|
| 1. 34%   | 5. 16%   | 9. 57%  |
| 2. 67.5% | 6. 72.5% | 10. 84% |
| 3. 95%   | 7. 35%   | 11. 10% |
| 4. 96%   | 8. 80%   | 12. 55% |

#### Challenging

1. 22%
2. Anglican
3. No religion

#### Extension

1. 76.5°
2. Gold Coast 6 cm, Uluru 3.4 cm, Snowy 3 cm, Kangaroo Is. 2 cm, Phillip Is. 1.6 cm

#### Exit ticket

1. Level 1: Phillip Island
2. Level 2: 11°C
3. Level 3: 8%

### 10.02 Misleading graphs (booklet p. 9)

#### WE DO examples (teacher)

1. Ex 1a: the vertical axis starts at \$2 million, not 0
2. Ex 1b: profits grew much faster than they did
3. Ex 2a: no
4. Ex 2b: 51 and 56
5. Ex 3a: uneven scale, pictures, no labels, missing data
6. Ex 3b: to impress investors

#### Easy (drill)

- |      |                   |                    |
|------|-------------------|--------------------|
| 1. M | 5. axis start     | 9. picture size    |
| 2. F | 6. picture size   | 10. axis start     |
| 3. F | 7. missing labels | 11. missing labels |
| 4. M | 8. scale          | 12. scale          |

#### Medium (drill)

- |                                      |                                                    |                                             |
|--------------------------------------|----------------------------------------------------|---------------------------------------------|
| 1. small differences look very large | 3. the change looks faster than it is              | 5. the front sector looks bigger than it is |
| 2. the missing months hide the falls | 4. its area is 4 times, so it looks 4 times as big |                                             |

- |                                                                          |                                    |                                     |
|--------------------------------------------------------------------------|------------------------------------|-------------------------------------|
| 6. the second column looks 3 times as tall, but sales only rose about 2% | 7. looks 4 times; really 1.1 times | 10. looks 5 times; really 1.4 times |
|                                                                          | 8. looks 5 times; really 1.1 times | 11. looks 4 times; really 1.3 times |
|                                                                          | 9. looks 2 times; really 1.0 times | 12. looks 3 times; really 1.0 times |

#### Challenging

1. the area is 4 times bigger, not 2
2. you do not know what is shown
3. the steps are not equal

#### Extension

1. the axis starts at 2, so heights above 2 are compared (0.2 vs 2.0)
2. e.g. axis from 0 to 20

#### Exit ticket

1. Level 1: e.g. the axis does not start at 0
2. Level 2: small differences look huge
3. Level 3: start the vertical axis at 0 with an even scale

### 10.03 Dot plots (booklet p. 15)

#### WE DO examples (teacher)

1. Ex 1a: 23°C
2. Ex 1b: 15°C
3. Ex 2a: 4
4. Ex 2b:  $\frac{1}{15}$
5. Ex 3a: dot plot
6. Ex 3b: most days between 20°C and 26°C, peak at 23°C

#### Easy (drill)

- |      |        |        |
|------|--------|--------|
| 1. 1 | 6. 8   | 11. 5  |
| 2. 2 | 7. 7   | 12. 8  |
| 3. 8 | 8. 2   | 13. 3  |
| 4. 6 | 9. 7   | 14. 11 |
| 5. 6 | 10. 18 | 15. 7  |

#### Medium (drill)

- |       |        |        |
|-------|--------|--------|
| 1. 36 | 7. 31  | 13. 37 |
| 2. 32 | 8. 13  | 14. 30 |
| 3. 31 | 9. 33  | 15. 22 |
| 4. 25 | 10. 34 | 16. 29 |
| 5. 29 | 11. 24 |        |
| 6. 26 | 12. 33 |        |

#### Challenging

1. dot plot
2. mode 0, outlier 15; it makes the mean bigger

#### Extension

1. 18.8%
2. still 1

#### Exit ticket

1. Level 1: a graph with one dot for each value on a number line
2. Level 2: 1
3. Level 3: 8

### 10.04 Stem-and-leaf plots (booklet p. 21)

#### WE DO examples (teacher)

1. Ex 1a: 25
2. Ex 1b: 71 kg
3. Ex 2a: 80 kg

4. Ex 2b: 106 kg
5. Ex 3a: 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
6. Ex 3b: to see the shape and find the median and mode easily

#### Easy (drill)

- |       |        |        |
|-------|--------|--------|
| 1. 47 | 6. 65  | 11. 65 |
| 2. 13 | 7. 27  | 12. 3  |
| 3. 11 | 8. 79  | 13. 9  |
| 4. 39 | 9. 16  | 14. 44 |
| 5. 12 | 10. 21 |        |

#### Medium (drill)

- |                   |                   |                    |
|-------------------|-------------------|--------------------|
| 1. 12, 15, 15, 18 | 5. 50, 51, 52, 56 | 9. 55, 55, 58, 59  |
| 2. 31, 31, 33, 38 | 6. 15, 15, 18, 19 | 10. 80, 82, 88, 89 |
| 3. 74, 74, 78, 79 | 7. 30, 31, 34, 36 | 11. 60, 62, 64, 65 |
| 4. 21, 24, 26, 29 | 8. 11, 12, 14, 14 | 12. 90, 94, 95, 99 |

#### Challenging

1. 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
2. 9 players, 9/25

#### Extension

1. 10 | 5; 11 | 2 8; 12 | 1 5; 13 | 0
2. 25 hundredths (0.25)

#### Exit ticket

1. Level 1: 6
2. Level 2: 40, 42, 47
3. Level 3: 1 | 2 4 7; 2 | 1 5 9

### 10.05 Mean and mode (booklet p. 27)

#### WE DO examples (teacher)

1. Ex 1a: 4
2. Ex 1b: 4
3. Ex 2a: 8.78
4. Ex 2b: 6
5. Ex 3a: 3.5; 1
6. Ex 3b: 6

#### Easy (drill)

- |       |        |        |
|-------|--------|--------|
| 1. 3  | 10. 12 | 19. 11 |
| 2. 11 | 11. 11 | 20. 1  |
| 3. 6  | 12. 11 | 21. 3  |
| 4. 10 | 13. 4  | 22. 10 |
| 5. 10 | 14. 10 | 23. 9  |
| 6. 7  | 15. 12 | 24. 1  |
| 7. 15 | 16. 12 | 25. 8  |
| 8. 9  | 17. 4  |        |
| 9. 10 | 18. 12 |        |

#### Medium (drill)

- |         |          |        |
|---------|----------|--------|
| 1. 31.4 | 8. 23    | 15. 6  |
| 2. 19.3 | 9. 33.5  | 16. 24 |
| 3. 30.7 | 10. 29   | 17. 20 |
| 4. 27.6 | 11. 12.3 | 18. 8  |
| 5. 27.3 | 12. 32   | 19. 8  |
| 6. 38.5 | 13. 40   | 20. 23 |
| 7. 28.1 | 14. 53   |        |

#### Challenging

1. 6
2. 84
3. e.g. 8, 8, 10, 11, 13

#### Extension

1. 10.4
2. it is a balance point, e.g. mean of 1 and 2 is 1.5

#### Exit ticket

1. Level 1: 5
2. Level 2: 8
3. Level 3: 6.5

### 10.06 Median and range (booklet p. 33)

#### WE DO examples (teacher)

1. Ex 1a: 4
2. Ex 1b: 4
3. Ex 2a: 10
4. Ex 2b: 6
5. Ex 3a: 3.5; 7
6. Ex 3b: the mean

#### Easy (drill)

- |       |        |        |
|-------|--------|--------|
| 1. 10 | 10. 19 | 19. 24 |
| 2. 12 | 11. 46 | 20. 32 |
| 3. 15 | 12. 47 | 21. 44 |
| 4. 12 | 13. 27 | 22. 35 |
| 5. 11 | 14. 45 | 23. 41 |
| 6. 22 | 15. 29 | 24. 37 |
| 7. 11 | 16. 41 | 25. 46 |
| 8. 20 | 17. 29 |        |
| 9. 13 | 18. 35 |        |

#### Medium (drill)

- |          |                           |                           |
|----------|---------------------------|---------------------------|
| 1. 23.5  | 14. 20.5                  | 21. median 47, range 50   |
| 2. 23    | 15. 6                     | 22. median 45, range 30   |
| 3. 17.5  | 16. median 22, range 43   | 23. median 35, range 40   |
| 4. 23    | 17. median 23, range 38   | 24. median 33, range 19   |
| 5. 15    | 18. median 29.5, range 40 | 25. median 47.5, range 31 |
| 6. 21    | 19. median 37, range 30   | 26. median 41, range 25   |
| 7. 18    | 20. median 43, range 32   | 27. median 36.5, range 46 |
| 8. 14.5  |                           |                           |
| 9. 19    |                           |                           |
| 10. 13   |                           |                           |
| 11. 28.5 |                           |                           |
| 12. 15   |                           |                           |
| 13. 26.5 |                           |                           |

#### Challenging

1. mean 7, median 3.5, mode 3, range 23
2. the median: the outlier 25 pulls the mean up
3. e.g. 1, 4, 6, 8, 11

#### Extension

1. 3, 5, 9, 9
2. a few very high salaries make the mean misleading

#### Exit ticket

1. Level 1: 6
2. Level 2: 25
3. Level 3: 9

### 10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 39)

#### WE DO examples (teacher)

1. Ex 1a: 7
2. Ex 1b: 1.5
3. Ex 2a: 66
4. Ex 2b: 64
5. Ex 3a: 61.2
6. Ex 3b: mostly in the 60s

#### Easy (drill)

1. 40

2. 27
3. 25
4. 11
5. no mode
6. 55
7. 40
8. 51.5
9. 5

#### Medium (drill)

- |        |                      |                       |
|--------|----------------------|-----------------------|
| 1. 14  | 7. 5                 | 12. 8                 |
| 2. 9   | 8. no (it becomes 7) | 13. 6.3               |
| 3. 6.5 | 9. 14                | 14. 1                 |
| 4. 8   | 10. 9                | 15. 5                 |
| 5. 6.3 | 11. 6.5              | 16. no (it becomes 7) |
| 6. 1   |                      |                       |

#### Challenging

1. 35
2. 83
3. 80

#### Extension

1. the median: it is not affected by extreme values
2. 1.53

#### Exit ticket

1. Level 1: 3
2. Level 2: 53.5
3. Level 3: 18

## 10.08 Comparing data (booklet p. 45)

#### WE DO examples (teacher)

1. Ex 1a: the marks of the girls (left) and boys (right)
2. Ex 1b: 94, a girl
3. Ex 2a: 54
4. Ex 2b: 83
5. Ex 3a: girls: mean 55, median 54; boys: mean 55.8, median 60

6. Ex 3b: girls: range 76 vs boys 83

#### Easy (drill)

- |                     |                        |                         |
|---------------------|------------------------|-------------------------|
| 1. A                | 8. A (8.75 and 7.25)   | 12. A (14 and 10.75)    |
| 2. B                | 9. B (6.75 and 10.25)  | 13. B (9.25 and 11.25)  |
| 3. A                | 10. A (15.25 and 5.75) | 14. B (14 and 15.5)     |
| 4. B                | 11. A (13.25 and 9.5)  | 15. A (7.75 and 5.25)   |
| 5. B                |                        | 16. A (11.75 and 11.25) |
| 6. B                |                        |                         |
| 7. A (13 and 10.75) |                        |                         |

#### Medium (drill)

- |                                                 |                                                 |                   |
|-------------------------------------------------|-------------------------------------------------|-------------------|
| 1. A mean 11, range 4; B mean 11.2, range 9     | 4. A mean 13.4, range 13; B mean 13.6, range 11 | 7. A 49, B 36: A  |
| 2. A mean 11.8, range 13; B mean 11.6, range 6  | 5. A mean 9.2, range 8; B mean 11.4, range 11   | 8. A 24, B 21: A  |
| 3. A mean 13.6, range 13; B mean 11.4, range 13 | 6. A mean 13, range 8; B mean 10.8, range 8     | 9. A 27, B 28: B  |
|                                                 |                                                 | 10. A 28, B 34: B |
|                                                 |                                                 | 11. A 51, B 36: A |
|                                                 |                                                 | 12. A 46, B 43: A |

#### Challenging

1. boys have a slightly higher median (60 vs 54) but a larger range
2. girls: smaller range (76 vs 83)
3. median, as there are some very low marks

#### Extension

1. Ana: she scores 12 or more every game
2. e.g. 5, 5, 5 and 0, 5, 10

#### Exit ticket

1. Level 1: comparing two groups
2. Level 2: A
3. Level 3: A